1. MONKEY BANANA PROBLEM

AIM

To write a **Prolog program to implement the Monkey and Banana problem** using facts and rules.

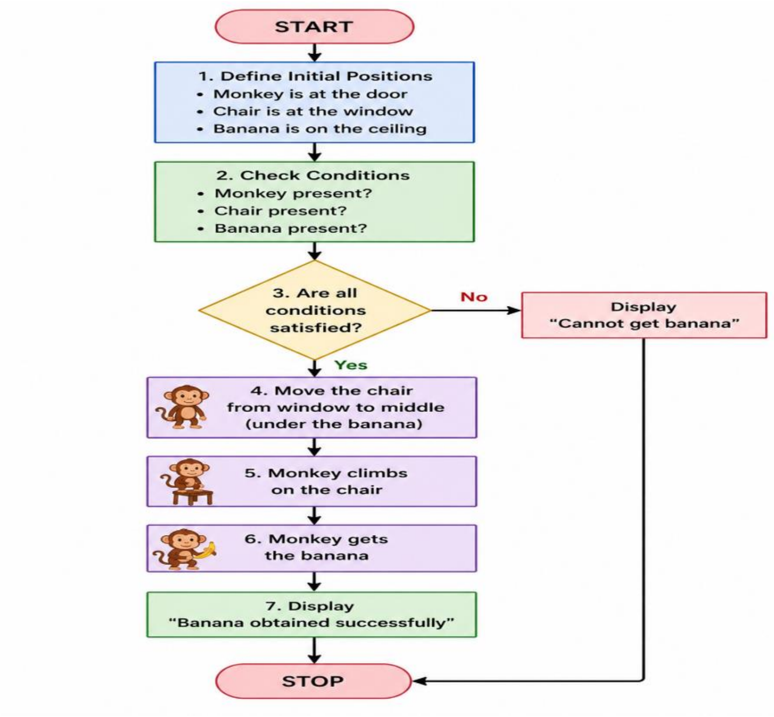
OBJECTIVE

- To represent the monkey, chair, and banana as facts.
- To use Prolog rules to solve the problem.
- To determine whether the monkey can obtain the banana.
- To understand knowledge representation and logical reasoning.

ALGORITHM

1. Start.
2. Define the positions of the monkey, chair, and banana.
3. Create a rule can_get_banana.
4. Check whether the required objects are present.
5. If the conditions are satisfied, perform the actions: move chair, climb, and get banana.
6. Display that the banana is obtained.
7. Stop.

FLOWCHART



CODE

```
can_get_banana :-
    move_chair,
    climb,
    get_banana.
move_chair :-
    write('Monkey moves chair, ').
climb :-
    write('climbs and ').
get_banana :-
    write('gets banana.'),
```

QUERY

```
?- can_get_banana.
```

nl.**OUTPUT**

```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
File   Edit   Settings   Run   Debug   Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/Users/New/Documents/Prolog/monkey banana.pl compiled 0.00 sec, 4 clauses
?-  can_get_banana.
Monkey moves chair, climbs and gets banana.
true.

?-
```

RESULT

Thus, the Monkey and Banana problem was successfully implemented in Prolog, and the monkey successfully obtains the banana when the required conditions are satisfied.